

Irradiation Effects of Fe-Cr Alloys

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ABSTRACT

Various researches have indicated the relative advantages of ferritic/martensitic (FM) steels over austenitic and other steels currently in use in light water reactors, and have regarded them as the most promising structural materials in both present and future reactors. Fe-Cr are the model alloys of these steels. Continuous exposure to severe irradiation environment will no doubt affect the microstructure of these materials significantly, which can subsequently manifest as observable changes in the materials' physical properties. Various experimental and simulation literatures were reviewed in this paper with the aim of further understanding the irradiation effects of Fe-Cr alloys, which are the reference model systems for high chromium steels. Factors such as; solute type and concentration, irradiation dose, temperature, number of displacements per atom, dpa (fluence/dose), as well as type and energy of irradiating particle, among other things, affect the production and evolution of radiation damage in Fe-Cr alloys. Researchers' findings on the way some of these factors affect the alloy under irradiation condition are presented in the paper. While the addition of Chromium (Cr) was found to improve the performance of FM steels by strongly suppressing void swelling, minimizing the effect of increase in irradiation hardening with decrease in dose rate, and so on, the effect of its content on Fe-Cr alloys under irradiation condition is however not clearly understood by the

experimental studies, perhaps due to factors such as the non-monotonic variation of its content with properties like void swelling. Primary radiation damage production and evolution, as presented in various Molecular Dynamics (MD) simulation literatures reviewed, determine the macroscopic response of a material to irradiation, thus making it paramount to understand. Insight into the atomic processes leading to changes in mechanical properties of materials can be gained through multi-scale computer simulations. Rigorous efforts are therefore needed in this regard to help enhance our understanding of the effects of irradiation on materials and the way they can best be mitigated during design in order to help ensure safe and reliable operation of nuclear power plants.

Keywords: Irradiation, Fe-Cr, chromium, alloys, simulation.

1. INTRODUCTION

Fe-Cr alloys are considered very suitable for use in both current and future reactors due to their high thermal conductivity, lower coefficient of thermal expansion, better void swelling resistance and high resistance to corrosion [1]–[5]. However, materials degradation, associated with neutron irradiation damage in nuclear fission reactors where fission of uranium nuclei by neutrons takes place, is a key issue. Therefore, despite the good properties of Fe-Cr alloys, radiation still causes